UNIVERSITY OF HERTFORDSHIRE

Assignment 1: LLM Coding Case Study and Report

AI-Assisted Financial Stability Analysis of Barbados (2024–2025)

PDF Processing, LLM-Based Information Extraction, Retrieval-Augmented Generation and Financial Data Visualisation

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1. Introduction

Large Language Models (LLMs) and artificial intelligence techniques provide opportunities to extract useful economic information from lengthy financial documents. However, automated analysis can produce inaccurate numerical values, incomplete summaries and unsupported interpretations. Therefore, reliable document processing, information retrieval and output evaluation are important.

This case study examines Barbados using the International Monetary Fund (IMF) Country Report No. 25/153, published in June 2025. The report discusses economic growth, inflation, public debt, banking stability and risks affecting the country's financial outlook (IMF, 2025).

The project developed a Python-based PDF-to-dashboard pipeline in Google Colab. It combined PDF text extraction, token-aware chunking, extractive summarisation, LLM-assisted thematic analysis, structured indicator extraction, FAISS-based retrieval and financial visualisation. The analysis also examined output reliability and limitations.

The objective was to transform unstructured IMF information into accessible economic indicators and visual insights while assessing the accuracy, traceability and practical limitations of AI-supported financial analysis.

2. Methodology and Pipeline Architecture

The project followed a four-stage PDF-to-dashboard pipeline developed in Python using Google Colab. The methodology combined document preprocessing, AI-assisted information extraction, retrieval-augmented generation (RAG) techniques and data visualisation. The source document was the International Monetary Fund's (IMF, 2025) Barbados Country Report No. 25/153.

2.1 PDF Processing and Summarisation

The 131-page IMF report was processed using PyMuPDF to extract textual information. The extracted content was cleaned to reduce formatting inconsistencies and divided into 297 overlapping chunks using the tiktoken tokenizer. Each chunk contained approximately 400 tokens with a 50-token overlap to preserve contextual continuity. TF-IDF-based extractive summarisation was applied to identify important sentences and produce executive and section summaries. This approach reduced computational requirements but provided less linguistic flexibility than generative LLM summarisation.

2.2 Financial Indicator Extraction and Evaluation

The second stage analysed seven financial stability themes and extracted economic indicators, national strengths and vulnerabilities. The Qwen2.5-0.5B-Instruct model supported structured information extraction and evaluation. Numerical outputs were stored in JSON format, with missing values represented as null. An automated judge assessed extracted information; however, the same model was used for extraction and evaluation, limiting assessment independence.

2.3 RAG Retrieval Architecture

A local retrieval system was developed using the all-MiniLM-L6-v2 sentence-transformer model to generate dense text embeddings. These were indexed using FAISS IndexFlatL2 for similarity-based retrieval (Johnson, Douze and Jégou,

2021). The corrected implementation combined FAISS with TF-IDF retrieval to improve relevant evidence identification. Retrieved passages supported source-grounded responses to financial questions.

2.4 Dashboard Development

Finally, Pandas transformed extracted indicators into long-format datasets using pd.melt. Four financial visualisations were generated and saved at 300 DPI. The pipeline prioritised local execution, reusable outputs and reduced API dependency, although extraction accuracy and incomplete numerical coverage remained important limitations.

3. Task 1: PDF Processing and Summarisation

3.1 Document Processing and Chunking

The IMF Barbados Country Report No. 25/153 was processed using Python in Google Colab. The document contained 131 PDF pages, which were converted into cleaned text and divided into 297 chunks. Token-aware chunking used a maximum of 400 tokens with a 50-token overlap to maintain contextual continuity. The processing stage completed in approximately 57.25 seconds, demonstrating efficient local execution without paid LLM API calls.

3.2 Summarisation Approach and Results

TF-IDF-based extractive summarisation identified relevant sentences for an executive summary and ten major report sections. Each section summary remained below the required 100-word limit. The extracted findings included Barbados' 4% GDP growth in 2024, the government's 4.4% primary surplus target and improvements in financial stability (IMF, 2025).

3.3 Evaluation and Critical Analysis

Automated checks assessed summary length, evidence similarity and unsupported numerical percentages. The evaluation recorded an evidence similarity score of 1.0 for the selected source sentences, indicating direct textual correspondence rather than independent factual accuracy. However, some summaries contained formatting errors, including incorrectly combined inflation figures and unrelated footnote text.

The approach avoided API quota limitations and reduced processing time, but it did not implement generative LLM summarisation or an LLM-as-a-Judge framework. Therefore, the results demonstrate successful extractive summarisation and basic validation rather than full compliance with the assignment's LLM evaluation requirements. Human verification remained necessary to confirm factual completeness and contextual accuracy.

4. Task 2: Thematic Extraction, Indicator Generation and Structured Output

4.1 Thematic Analysis and Economic Findings

Task 2 used Qwen2.5-0.5B-Instruct to analyse seven financial stability themes: inflation, interest rates and credit, housing, banking capital and liquidity, sovereign debt, labour markets, and systemic financial risk. The system generated seven economic strengths and seven vulnerabilities, supported by retrieved IMF passages.

Key strengths included tourism recovery, improving fiscal performance, stronger international reserves and declining unemployment. Major vulnerabilities concerned public debt, dependence on tourism, external economic shocks and climate-related disasters (IMF, 2025).

4.2 Numerical Indicator Extraction

The system attempted to extract six numerical indicators into a predefined JSON schema. However, all six numerical fields remained null in the original required_indicators.json, although supporting evidence was retrieved. This demonstrates that evidence identification did not successfully translate into validated numerical extraction.

The IMF report provides a 2025 GDP growth forecast of 2.7%, 2024 average inflation of 1.4% and unemployment of 7.1% in Q3 2024 (IMF, 2025). These values require separate verification before inclusion in the structured dataset. The

missing benchmark policy rate reflects an important reporting limitation because the Central Bank was still exploring its introduction.

4.3 LLM Evaluation and Limitations

The automated evaluation examined thematic responses against retrieved evidence and identified potentially unsupported percentages. However, the same Qwen model performed extraction and judging, limiting evaluation independence.

Some outputs also contained factual inconsistencies. For example, the inflation analysis incorrectly described sustained high inflation, while the IMF reported moderating inflation. Another vulnerability incorrectly located Barbados on Hispaniola.

Overall, Task 2 demonstrated thematic information retrieval and JSON generation, but numerical completeness and factual reliability remained limited. Independent evaluation and manual source verification were necessary.

5. Task 3: RAG Grounding and Retrieval

5.1 RAG Implementation

Task 3 implemented a local Retrieval-Augmented Generation (RAG) retrieval pipeline to improve factual grounding in the IMF Barbados report. The system converted 297 document chunks into dense vector embeddings using all-MiniLM-L6-v2. These embeddings were stored in FAISS IndexFlatL2, which supports similarity-based document retrieval (Johnson, Douze and Jégou, 2021).

The corrected implementation combined FAISS semantic search with TF-IDF keyword retrieval. This hybrid approach aimed to improve the identification of relevant financial information, particularly numerical indicators and economic terminology.

5.2 Retrieval Results and Evaluation

The system evaluated ten factual questions covering GDP growth, inflation, unemployment, public debt, banking conditions and economic risks. Four answers were classified as source-verified, three as extractive, two as qualified and one as abstained.

The corrected outputs identified 2025 GDP growth of 2.7%, unemployment of 7.1% in Q3 2024 and a decline in public debt from 111.5% to 104.8% of GDP (IMF, 2025). Source-page citations improved the traceability of retrieved evidence.

5.3 Critical Evaluation

The corrected pipeline completed in approximately 16.6 seconds. Compared with earlier responses, it reduced unsupported numerical claims through evidence checks and explicit abstention.

However, the correction used source-gated templates and extractive answers rather than generative LLM responses. Some answers were incomplete, particularly those concerning financial risks and natural disasters. Furthermore, the commercial-bank NPL query returned insufficient evidence. Therefore, the system demonstrated functional hybrid retrieval, but not comprehensive generative RAG evaluation or independently measured retrieval accuracy.

6. Task 4: Dashboard and Financial Insights

6.1 Data Transformation and Dashboard Development

Task 4 transformed the extracted IMF financial indicators into a visual dashboard using Python, Pandas and Matplotlib. The pd.melt() function converted wide-format economic indicators into long-format datasets, improving their suitability for comparison and visualisation. Missing numerical values were preserved rather than replaced with zero. Four charts were generated at 300 DPI, meeting the assignment's visualisation requirements.

6.2 Dashboard Results and Economic Interpretation

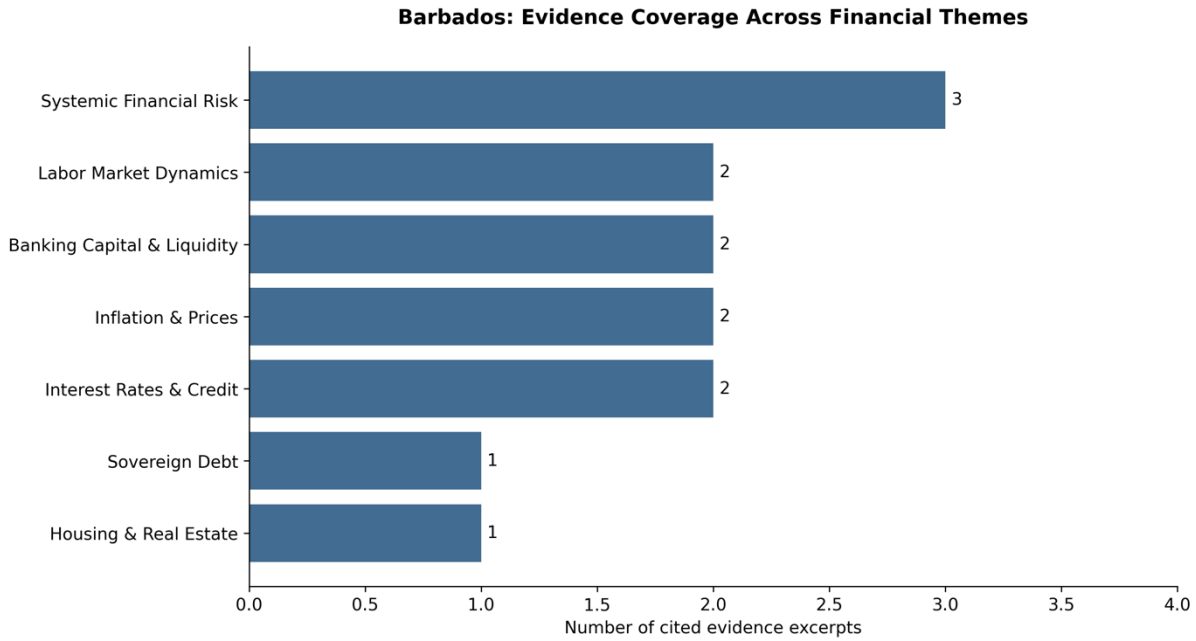


Figure 1: Financial Theme Distribution presents evidence-reference counts across seven financial stability themes. Systemic financial risk received three references, while housing and sovereign debt received one each. However, these counts represent documented evidence rather than the economic severity of each risk.

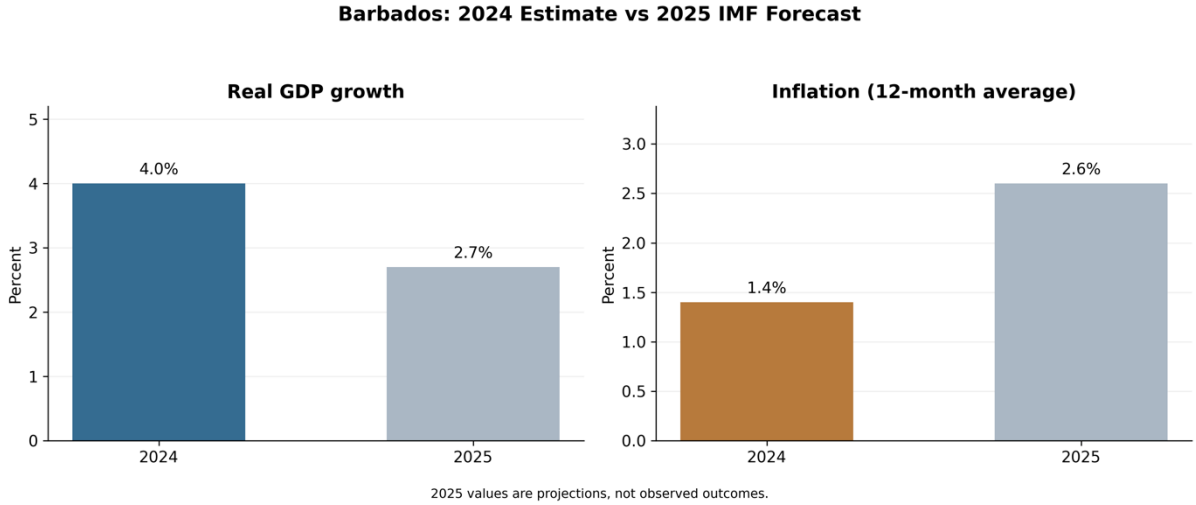


Figure 2: GDP Growth and Inflation Trends shows Barbados' GDP growth declining from 4.0% in 2024 to a projected 2.7% in 2025. Average inflation is expected to increase from 1.4% to 2.6% over the same period (IMF, 2025). This indicates slower projected economic expansion alongside renewed price pressures.

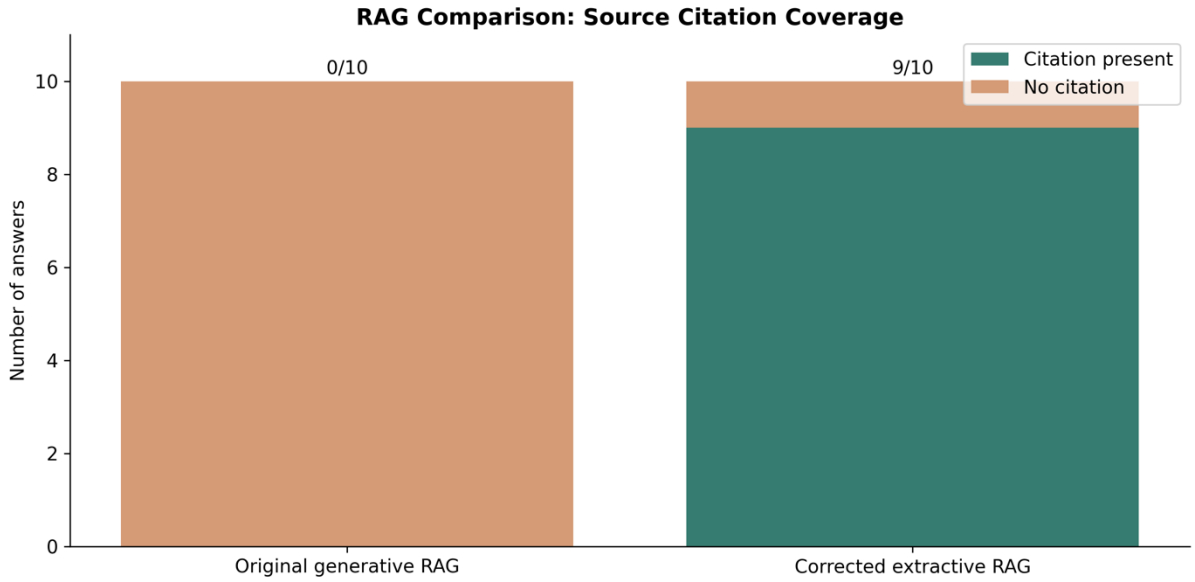


Figure 3: RAG Citation Comparison demonstrates improved source traceability. The original retrieval responses contained no page citations, whereas nine of ten corrected responses included citations. Nevertheless, citation coverage does not establish factual accuracy.

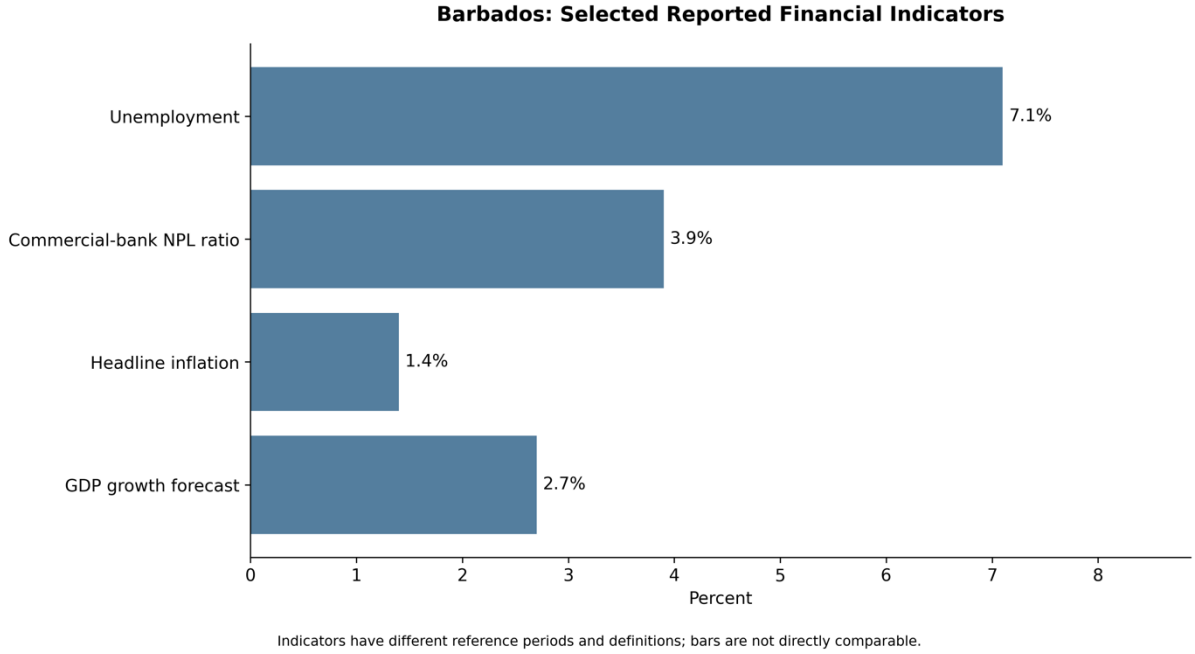


Figure 4: Selected Financial Indicators displays four available indicators, including GDP growth, inflation, commercial-bank non-performing loans (3.9%) and unemployment (7.1%). The policy rate and Tier 1 capital ratio remained unavailable.

6.3 Critical Financial Insights

The dashboard highlights Barbados' economic recovery alongside continuing vulnerabilities. Although banking conditions and public debt have improved, external shocks, tourism dependence and natural disasters remain significant concerns (IMF, 2025). Interpretation is limited by differing reporting periods, incomplete indicators and the small number of observations. Therefore, the dashboard provides descriptive insights rather than statistically validated forecasts.

7. Critical Evaluation and Limitations

The developed pipeline successfully processed the IMF report, generated thematic outputs, implemented vector-based retrieval and produced four dashboard visualisations. However, several limitations affected reliability.

Task 1 relied on TF-IDF extractive summarisation rather than generative LLMs, resulting in occasional formatting errors and incomplete contextual interpretation. Task 2 produced six null numerical indicators, demonstrating weaknesses in schema-constrained extraction. The Qwen model also generated factual inaccuracies, including misleading inflation interpretations and geographical errors. Using the same model for extraction and evaluation reduced judge independence.

Task 3 improved evidence traceability through hybrid FAISS and TF-IDF retrieval, but citation availability did not guarantee factual correctness. Task 4 was limited by missing indicators and inconsistent reporting periods.

Future improvements should include stronger LLM models, independent evaluation, stricter numerical validation, improved PDF table extraction and quantitative retrieval benchmarking.

8. Conclusion

This case study demonstrated how Python-based AI techniques can transform a lengthy IMF report into structured financial information and visual insights. The pipeline processed 131 pages, analysed seven financial themes, implemented hybrid FAISS retrieval and generated four dashboard charts. However, incomplete numerical extraction, occasional hallucinations and limited evaluation independence affected reliability. Overall, the findings demonstrate the value of AI-assisted financial analysis while emphasising the importance of evidence verification and human oversight.

9. References

International Monetary Fund (IMF) (2025) Barbados: Fifth Reviews Under the Extended Arrangement Under the Extended Fund Facility and the Arrangement Under the Resilience and Sustainability Facility. IMF Country Report No. 25/153. Washington, DC: International Monetary Fund. Available at: https://www.imf.org/en/Publications/CR/Issues/2025/06/ (Accessed: 9 November 2026).

Johnson, J., Douze, M. and Jégou, H. (2021) 'Billion-scale similarity search with GPUs', IEEE Transactions on Big Data, 7(3), pp. 535–547. Available at: https://doi.org/10.1109/TBDATA.2019.2921572.